

Lab Title: Case Study 3 — Illegal File Transferring: Memory, USB and User-Activity Forensics

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Student ID: 2026/DFIT/17036

Course Name: Computer Forensics Case Study II (CIP-B105)

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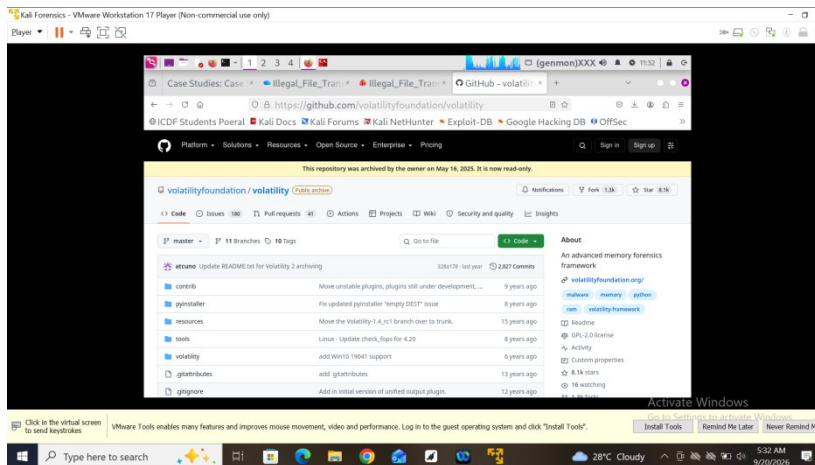
Instruction Name: Aminu Idris

Platform Used: ICDFA Kali Linux

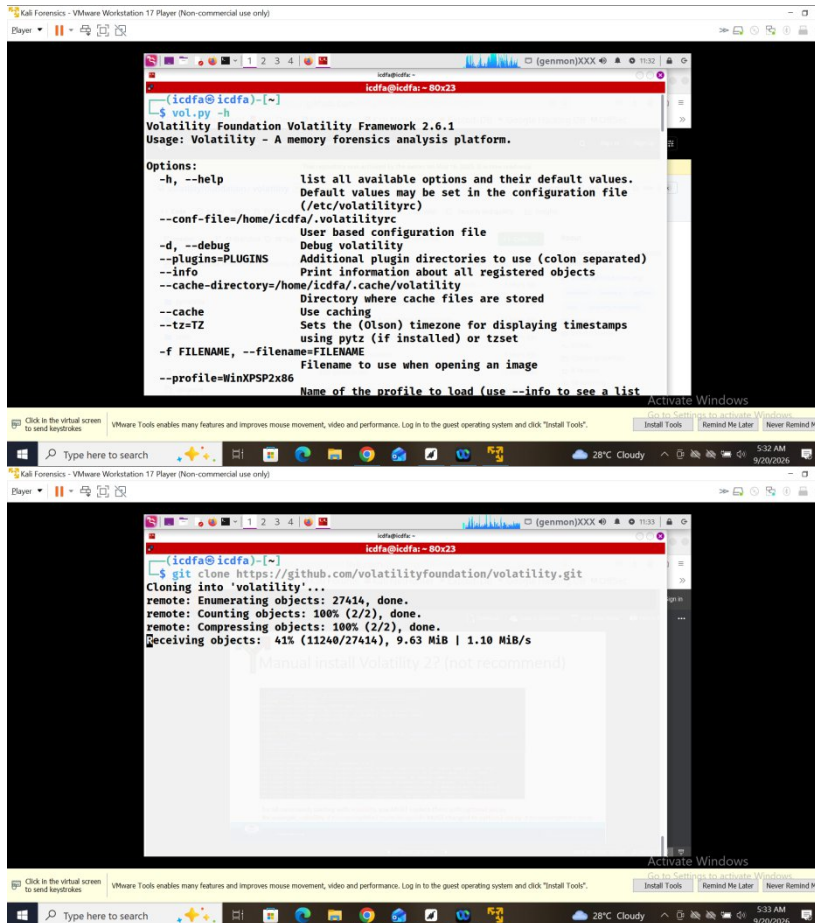
Executive Summary

The Volatility 2 memory analysis identified the suspect account and PC and correlated browser, process, command-history, Shellbag, and USB artefacts. The evidence shows a sequence consistent with downloading a file, accessing the Downloads folder, connecting a removable USB device, and copying the file to the assigned removable drive. USB Registry artefacts confirmed the device identity, drive-letter assignment, and user association. These findings were corroborated and aligned in a corrected UTC timeline.

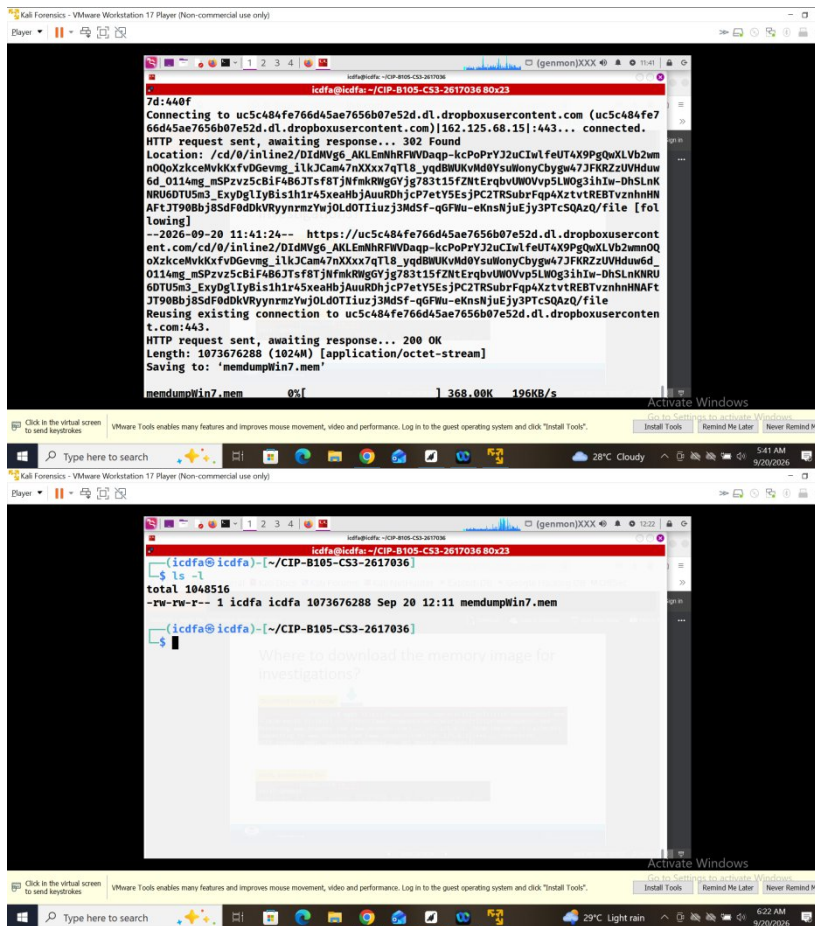
1. The Github site for volatility



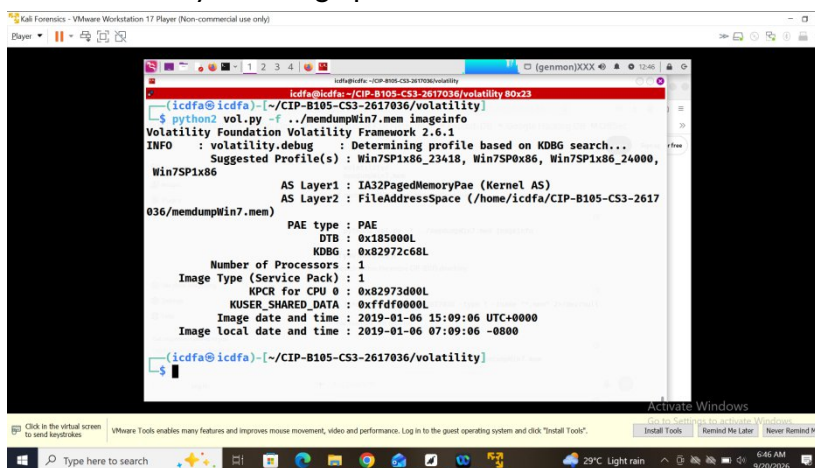
2. How to install volatility



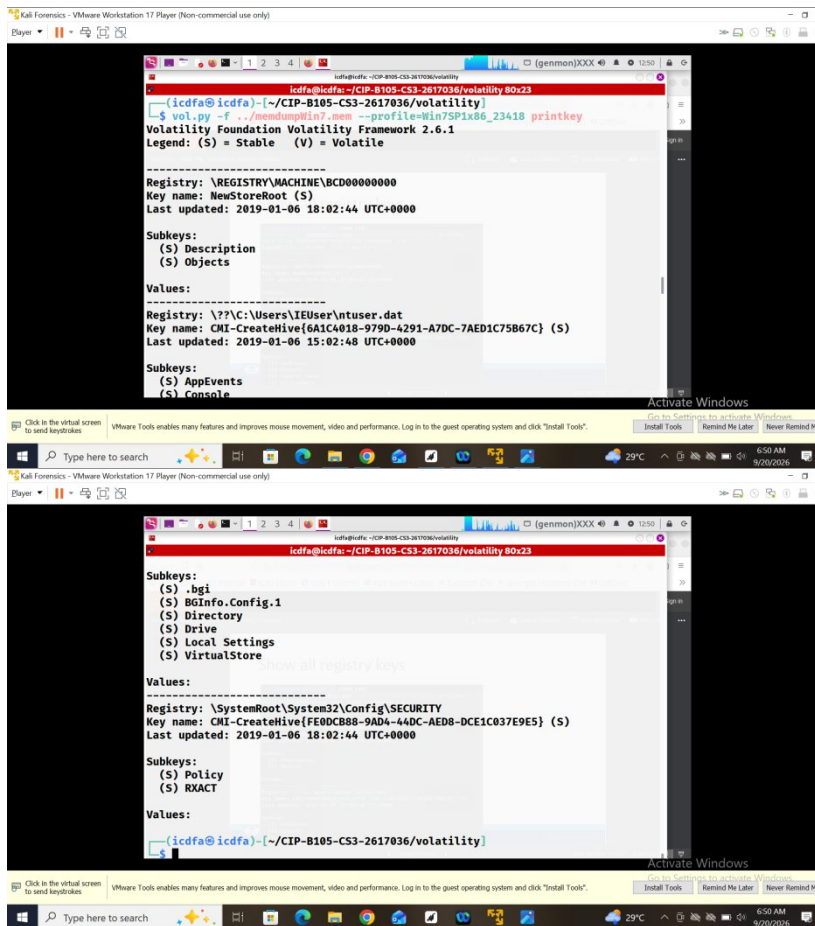
3. Downloading the memory image for investigation and verifying the download



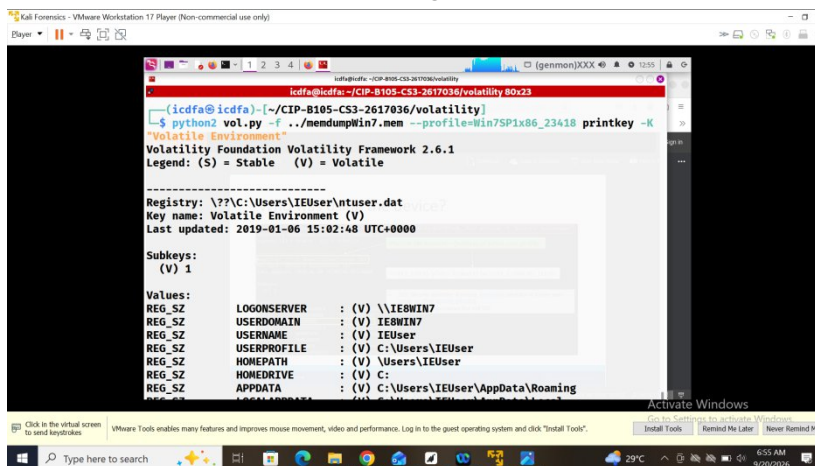
4. How to identify the image profile



5. Show all registry keys



6. Here was to check who was using the device



7. Who are associated with the device

```
icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility 80x23
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable (V) = Volatile

Registry: \SystemRoot\System32\Config\SOFTWARE
Key name: ProfileList (S)
Last updated: 2015-09-21 09:50:52 UTC+0000

Subkeys:
(S) S-1-5-18
(S) S-1-5-19
(S) S-1-5-20
(S) S-1-5-21-1716914095-909560446-1177810406-1000
(S) S-1-5-21-1716914095-909560446-1177810406-1002

Values:
REG_EXPAND_SZ ProfilesDirectory : (S) %SystemDrive%\Users
REG_EXPAND_SZ Default : (S) %SystemDrive%\Users\Default
REG_EXPAND_SZ Public : (S) %SystemDrive%\Users\Public
REG_EXPAND_SZ ProgramData : (S) %SystemDrive%\ProgramData

icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility
```

8. Who has a particular SID

```
icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility 80x23
Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1002
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable (V) = Volatile

Registry: \SystemRoot\System32\Config\SOFTWARE
Key name: S-1-5-21-1716914095-909560446-1177810406-1002 (S)
Last updated: 2015-09-21 09:51:49 UTC+0000

Subkeys:

Values:
REG_EXPAND_SZ ProfileImagePath : (S) C:\Users\sshd_server
REG_DWORD Flags : (S) 0
REG_DWORD State : (S) 256
REG_BINARY Sid : (S)
0x00000000 01 05 00 00 00 00 05 15 00 00 00 af 07 56 66 .....VF
0x00000010 7e ca 36 e6 f5 33 46 ea 03 00 00 .....66..3F....
REG_DWORD ProfileLoadTimeLow : (S) 0
REG_DWORD ProfileLoadTimeHigh : (S) 0
REG_DWORD RefCount : (S) 1
```

9. Checking who has the second SID as well

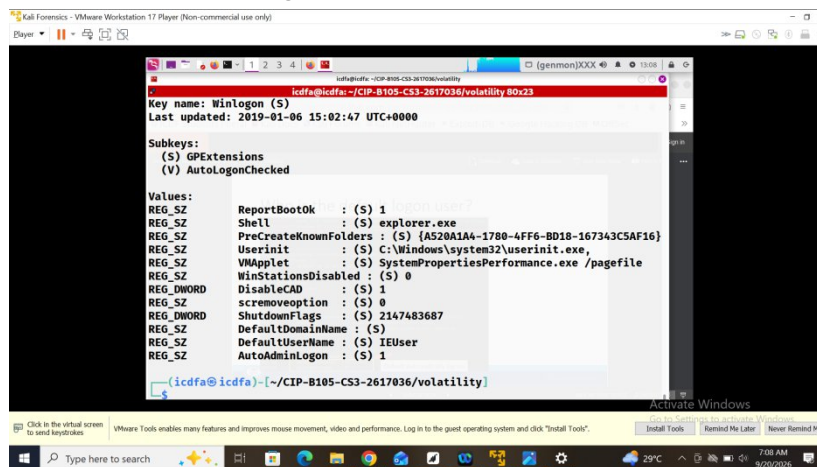
```
icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility 80x23
177810406-1000
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable (V) = Volatile

Registry: \SystemRoot\System32\Config\SOFTWARE
Key name: S-1-5-21-1716914095-909560446-1177810406-1000 (S)
Last updated: 2019-01-06 15:07:15 UTC+0000

Subkeys:

Values:
REG_EXPAND_SZ ProfileImagePath : (S) C:\Users\IEUser
REG_DWORD Flags : (S) 0
REG_DWORD State : (S) 0
REG_BINARY Sid : (S)
0x00000000 01 05 00 00 00 00 05 15 00 00 00 af 07 56 66 .....VF
0x00000010 7e ca 36 e6 f5 33 46 ea 03 00 00 .....66..3F....
REG_DWORD ProfileLoadTimeLow : (S) 0
REG_DWORD ProfileLoadTimeHigh : (S) 0
REG_DWORD RefCount : (S) 3
REG_DWORD RunLogonScriptsSync : (S) 0
```

10. Who is the default logon user



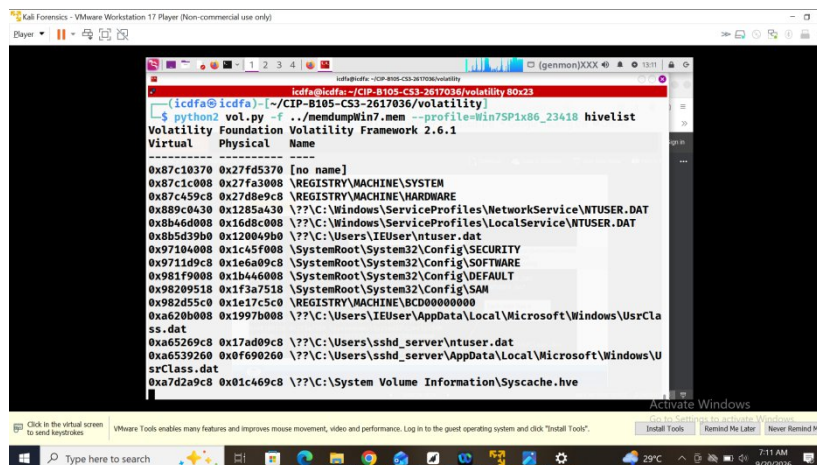
```
icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility 80x23
Key name: Winlogon (S)
Last updated: 2019-01-06 15:02:47 UTC+0000

Subkeys:
(S) GPExtensions
(V) AutoLogonChecked

Values:
REG_SZ ReportBootOk : (S) 1
REG_SZ Shell : (S) explorer.exe
REG_SZ PreCreateKnownFolders : (S) {A520A1A4-1780-4FF6-BD18-167343C5AF16}
REG_SZ Userinit : (S) C:\Windows\system32\userinit.exe,
REG_SZ VMApplet : (S) SystemPropertiesPerformance.exe /pagefile
REG_SZ WinStationsDisabled : (S) 0
REG_DWORD DisableCAD : (S) 1
REG_SZ Screenshotption : (S) 0
REG_DWORD ShutdownFlags : (S) 2147483687
REG_SZ DefaultDomainName : (S)
REG_SZ DefaultUserName : (S) IEUser
REG_SZ AutoAdminLogon : (S) 1

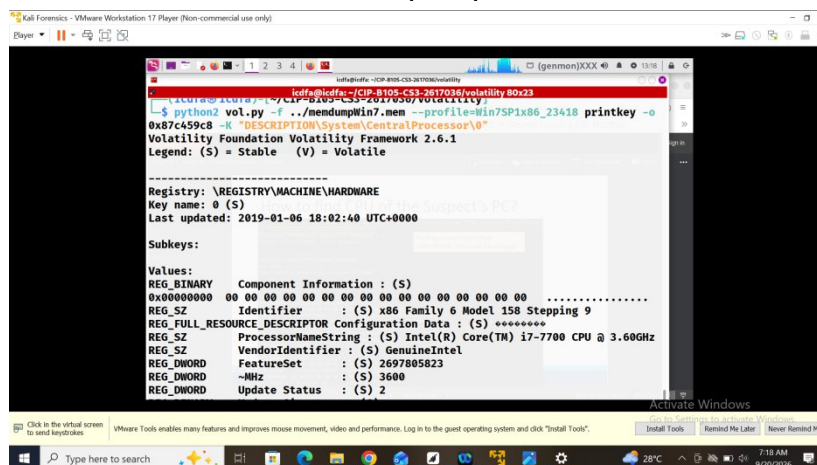
icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility
```

11. Here is the Hive overview



```
icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility 80x23
icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility
$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86_23418 hivelist
Volatility Foundation Volatility Framework 2.6.1
Virtual Physical Name
-----
0x87c10370 0x27fd5370 [no name]
0x87c1c008 0x27fa3008 \REGISTRY\MACHINE\SYSTEM
0x87c459c8 0x27d8e9c8 \REGISTRY\MACHINE\HARDWARE
0x889c0430 0x1285a430 \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT
0x8b46d008 0x16d8c008 \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT
0x8b5d3900 0x12004900 \??\C:\Users\IEUser\ntuser.dat
0x97104008 0x1c45f008 \SystemRoot\System32\Config\SECURITY
0x9711d9c8 0x1e6a09c8 \SystemRoot\System32\Config\SOFTWARE
0x981f9008 0x1b446008 \SystemRoot\System32\Config\DEFAULT
0x98209518 0x1f3a7518 \SystemRoot\System32\Config\SAM
0x982d55c0 0x1e17c5c0 \REGISTRY\MACHINE\BCD00000000
0xa6520b08 0x1997b008 \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa65269c8 0x17ad09c8 \??\C:\Users\sshd_server\ntuser.dat
0xa6539260 0x0f690260 \??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa7d2a9c8 0x01c469c8 \??\C:\System Volume Information\Syscache.hve
```

12. How to find the CPU of the suspect pc

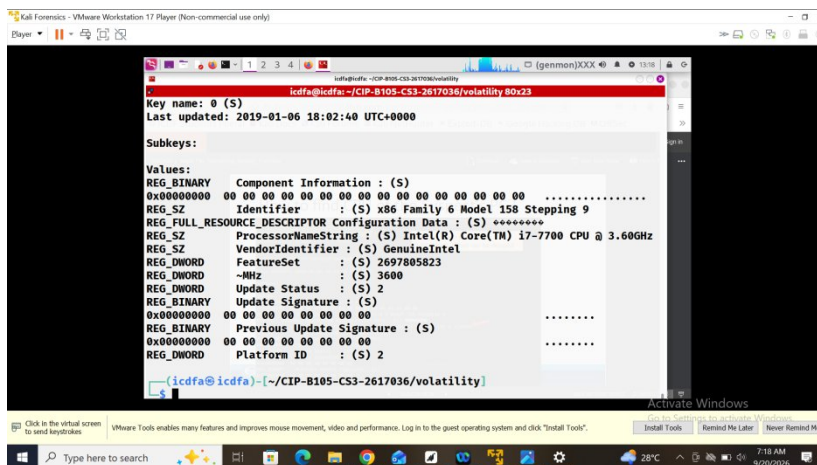


```
icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility 80x23
icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility
$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -o 0x87c459c8 -k "\REGISTRY\MACHINE\HARDWARE\ProcessorNameString"
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable (V) = Volatile

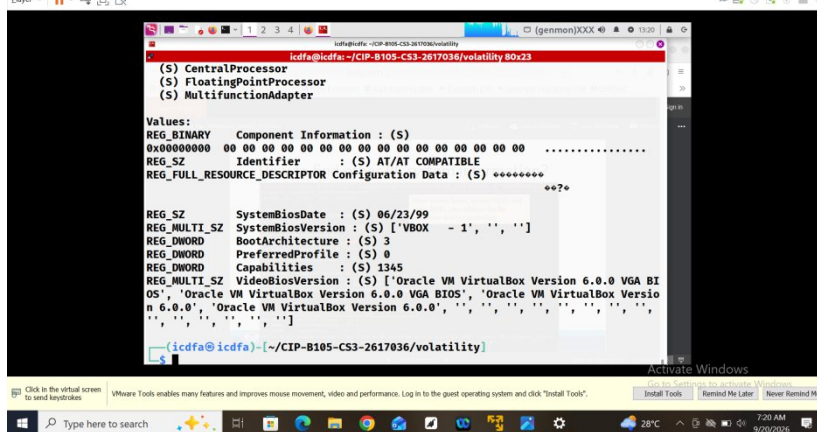
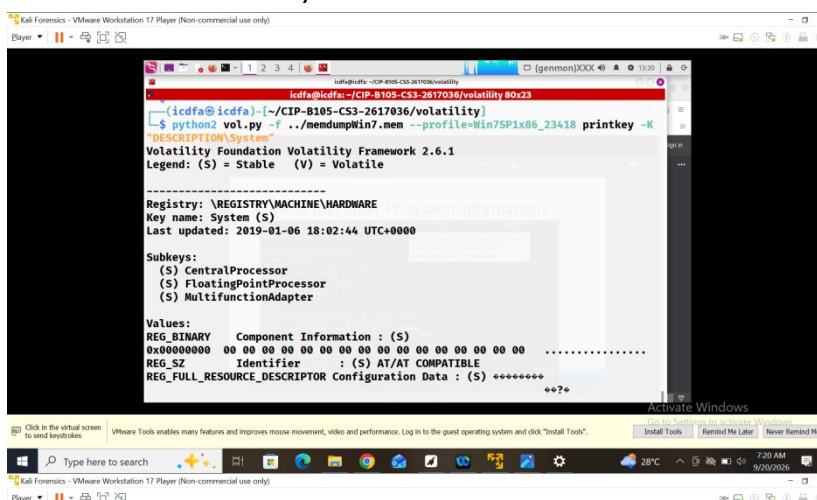
Registry: \REGISTRY\MACHINE\HARDWARE
Key name: 0 (S)
Last updated: 2019-01-06 18:02:40 UTC+0000

Subkeys:

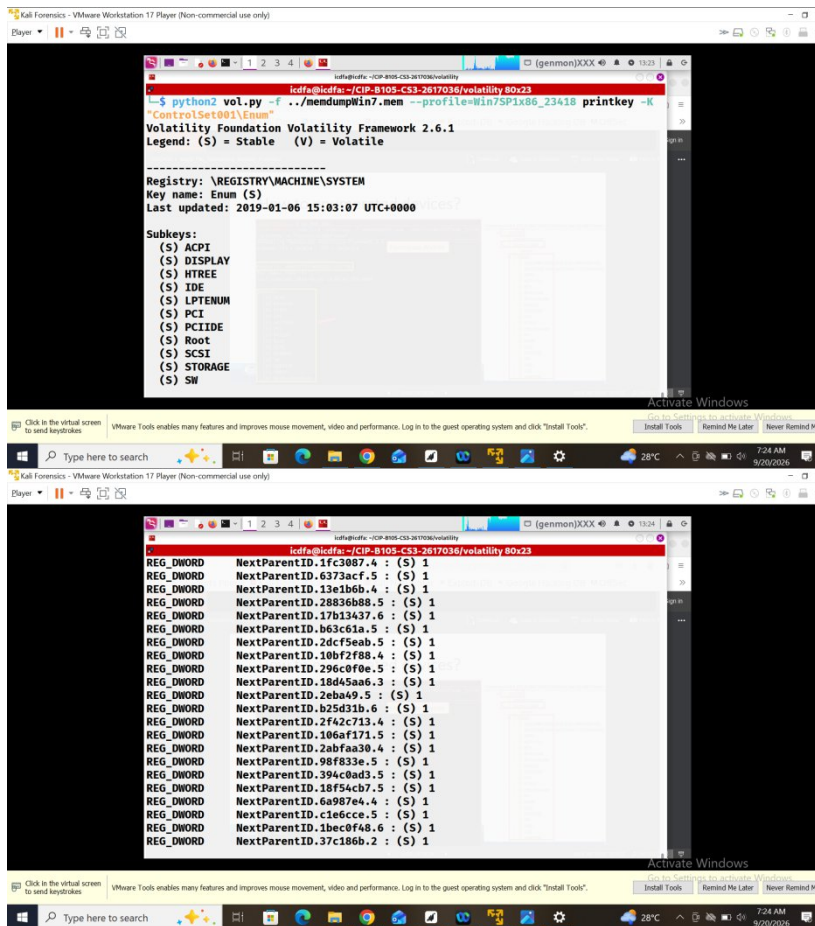
Values:
REG_BINARY Component Information : (S)
0x00000000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
REG_SZ Identifier : (S) x86 Family 6 Model 158 Stepping 9
REG_FULL_RESOURCE_DESCRIPTOR Configuration Data : (S) *****
REG_SZ ProcessorNameString : (S) Intel(R) Core(TM) i7-7700 CPU @ 3.60GHz
REG_SZ VendorIdentifier : (S) GenuineIntel
REG_DWORD FeatureSet : (S) 2697805823
REG_DWORD ~Mhz : (S) 3600
REG_DWORD Update Status : (S) 2
```



13. How to find other PC system information



14. Enumerating devices



The screenshot shows a terminal window in a Kali Linux virtual machine. The command executed is `python2 vol.py -f ../mendumWin7.mem --profile=Win7SP1x86_23418 printkey -K "ControlSet001\Enum"`. The output displays the registry path `\REGISTRY\MACHINE\SYSTEM` and the key name `Enum (S)`. It lists subkeys including `ACPI`, `DISPLAY`, `HTREE`, `IDE`, `LPTENUM`, `PCI`, `PCIIDE`, `Root`, `SCSI`, `STORAGE`, and `SN`. The output also shows a list of registry values with their NextParentID and status (S for Stable, V for Volatile).

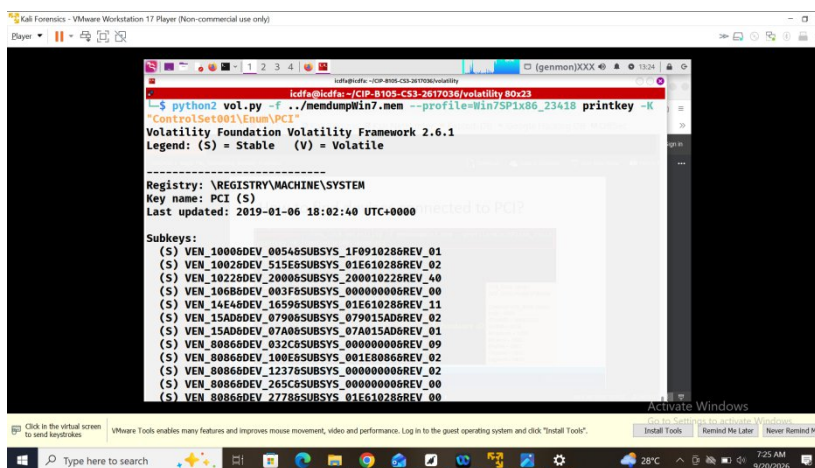
```
idfa@icdfa:~/CIP-B105-CS3-2617036/volatility BOx23
$ python2 vol.py -f ../mendumWin7.mem --profile=Win7SP1x86_23418 printkey -K
"ControlSet001\Enum"
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable (V) = Volatile

Registry: \REGISTRY\MACHINE\SYSTEM
Key name: Enum (S)
Last updated: 2019-01-06 15:03:07 UTC+0000

Subkeys:
(S) ACPI
(S) DISPLAY
(S) HTREE
(S) IDE
(S) LPTENUM
(S) PCI
(S) PCIIDE
(S) Root
(S) SCSI
(S) STORAGE
(S) SN

REG_DWORD NextParentID.1fc3087.4 : (S) 1
REG_DWORD NextParentID.6373acf.5 : (S) 1
REG_DWORD NextParentID.13e1b6b.4 : (S) 1
REG_DWORD NextParentID.28836b88.5 : (S) 1
REG_DWORD NextParentID.17b13437.6 : (S) 1
REG_DWORD NextParentID.b63c61a.5 : (S) 1
REG_DWORD NextParentID.2dcf5eab.5 : (S) 1
REG_DWORD NextParentID.10bf2f88.4 : (S) 1
REG_DWORD NextParentID.296c0f0e.5 : (S) 1
REG_DWORD NextParentID.18d45aa6.3 : (S) 1
REG_DWORD NextParentID.2eba49.5 : (S) 1
REG_DWORD NextParentID.b25d31b.6 : (S) 1
REG_DWORD NextParentID.2f42c713.4 : (S) 1
REG_DWORD NextParentID.106af171.5 : (S) 1
REG_DWORD NextParentID.2abfaa30.4 : (S) 1
REG_DWORD NextParentID.98f833e.5 : (S) 1
REG_DWORD NextParentID.394c0ad3.5 : (S) 1
REG_DWORD NextParentID.18f54cb7.5 : (S) 1
REG_DWORD NextParentID.6a987e4.4 : (S) 1
REG_DWORD NextParentID.c1e6cce.5 : (S) 1
REG_DWORD NextParentID.1bec0f4b.6 : (S) 1
REG_DWORD NextParentID.37c186b.2 : (S) 1
```

15. Devices Connected to PCI

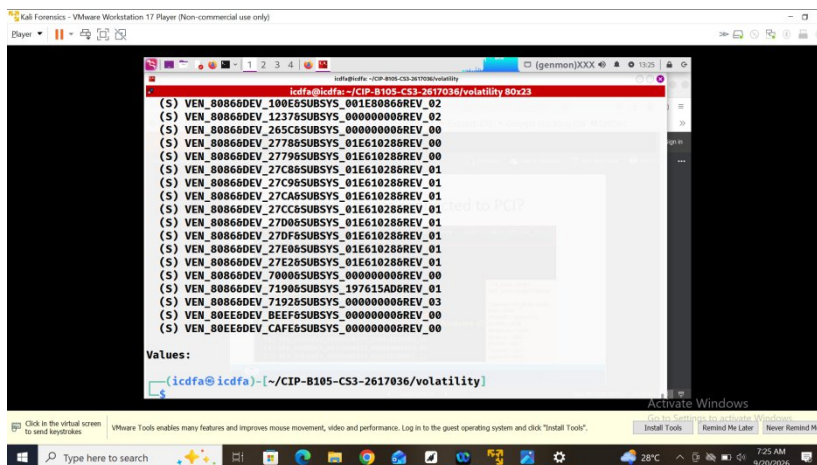


The screenshot shows a terminal window in a Kali Linux virtual machine. The command executed is `python2 vol.py -f ../mendumWin7.mem --profile=Win7SP1x86_23418 printkey -K "ControlSet001\Enum\PCI"`. The output displays the registry path `\REGISTRY\MACHINE\SYSTEM` and the key name `PCI (S)`. It lists subkeys including `VEN_10006DEV_00546SUBSYS_1F0910286REV_01`, `VEN_10026DEV_515E6SUBSYS_01E610286REV_02`, `VEN_10226DEV_20006SUBSYS_200010228REV_40`, `VEN_10686DEV_003F6SUBSYS_00000006REV_00`, `VEN_14E46DEV_16596SUBSYS_01E610286REV_11`, `VEN_15AD6DEV_07906SUBSYS_079015AD6REV_02`, `VEN_15AD6DEV_07A06SUBSYS_07A015AD6REV_01`, `VEN_80866DEV_032C6SUBSYS_00000006REV_09`, `VEN_80866DEV_100E6SUBSYS_001E80866REV_02`, `VEN_80866DEV_12376SUBSYS_00000006REV_02`, `VEN_80866DEV_265C6SUBSYS_00000006REV_00`, and `VEN_80866DEV_27785SUBSYS_01E610286REV_00`.

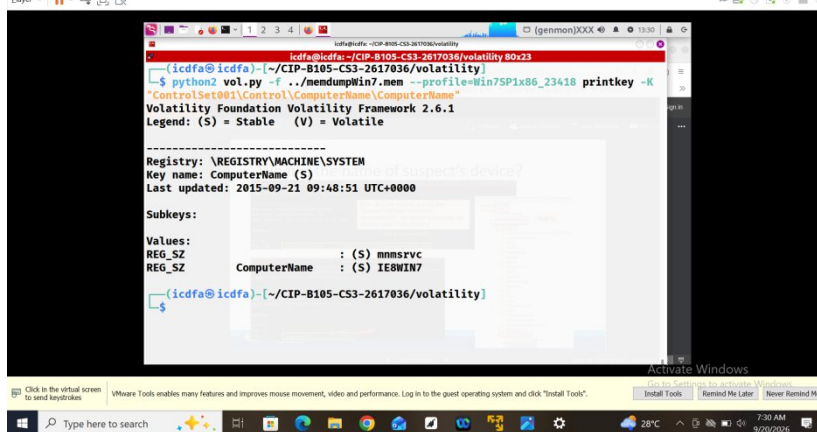
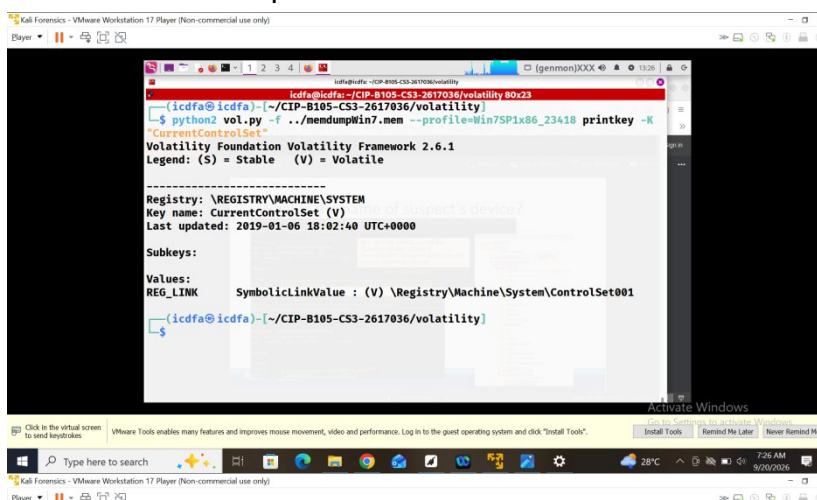
```
idfa@icdfa:~/CIP-B105-CS3-2617036/volatility BOx23
$ python2 vol.py -f ../mendumWin7.mem --profile=Win7SP1x86_23418 printkey -K
"ControlSet001\Enum\PCI"
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable (V) = Volatile

Registry: \REGISTRY\MACHINE\SYSTEM
Key name: PCI (S)
Last updated: 2019-01-06 18:02:40 UTC+0000

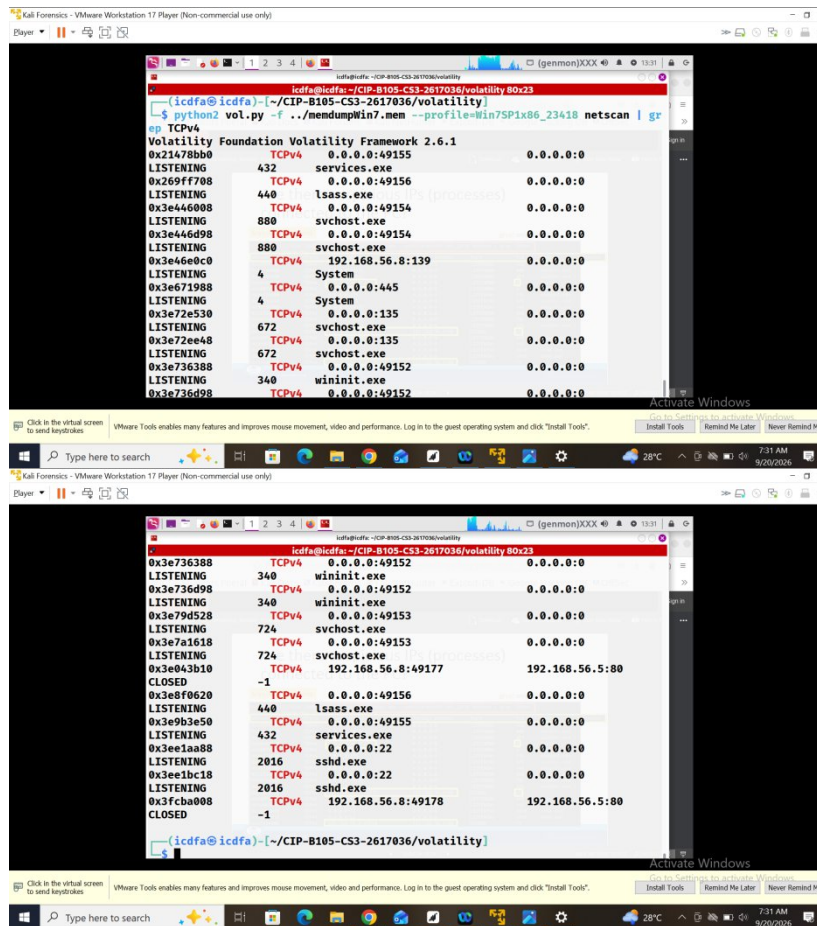
Subkeys:
(S) VEN_10006DEV_00546SUBSYS_1F0910286REV_01
(S) VEN_10026DEV_515E6SUBSYS_01E610286REV_02
(S) VEN_10226DEV_20006SUBSYS_200010228REV_40
(S) VEN_10686DEV_003F6SUBSYS_00000006REV_00
(S) VEN_14E46DEV_16596SUBSYS_01E610286REV_11
(S) VEN_15AD6DEV_07906SUBSYS_079015AD6REV_02
(S) VEN_15AD6DEV_07A06SUBSYS_07A015AD6REV_01
(S) VEN_80866DEV_032C6SUBSYS_00000006REV_09
(S) VEN_80866DEV_100E6SUBSYS_001E80866REV_02
(S) VEN_80866DEV_12376SUBSYS_00000006REV_02
(S) VEN_80866DEV_265C6SUBSYS_00000006REV_00
(S) VEN_80866DEV_27785SUBSYS_01E610286REV_00
```



16. The name of the suspect device



17. Suspicious IPs Connected to the PC



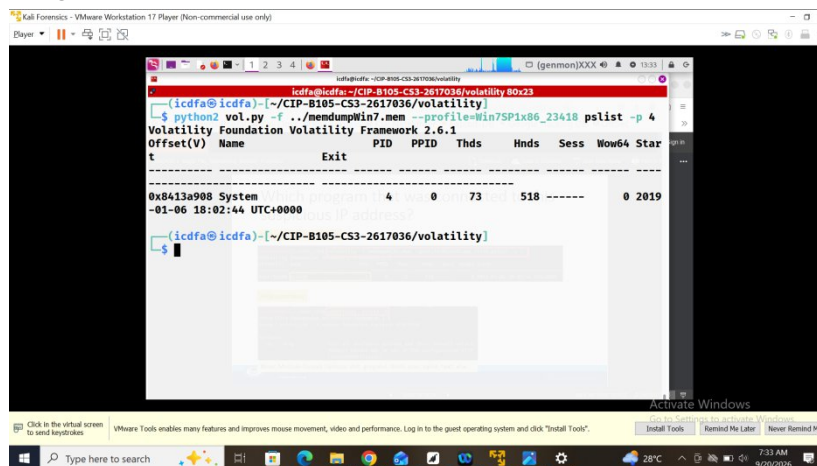
The first screenshot shows the output of the command `python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86_23418 netstat | grep TCPv4`. It displays a list of listening TCPv4 connections with their local addresses, ports, and the processes they are associated with.

Local Address	Port	Process	Remote Address
0x224770b00	432	services.exe	0.0.0.0:0
0x269ff708	440	lsass.exe	0.0.0.0:0
0x3e446008	880	svchost.exe	0.0.0.0:0
0x3e446d98	880	svchost.exe	0.0.0.0:0
0x3e46e0c0	4	System	192.168.56.8:139
0x3e671988	4	System	0.0.0.0:445
0x3e72e530	672	svchost.exe	0.0.0.0:135
0x3e72ee48	672	svchost.exe	0.0.0.0:135
0x3e736388	340	wininit.exe	0.0.0.0:49152
0x3e736d98	-1	wininit.exe	0.0.0.0:49152

The second screenshot shows the output of the command `python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86_23418 pslist -p 4`. It displays a list of processes with their PIDs, PPIDs, and other details.

PID	PPID	Process Name
0x3e736388	0.0.0.0:49152	wininit.exe
0x3e736d98	0.0.0.0:49152	wininit.exe
0x3e79d528	0.0.0.0:49153	svchost.exe
0x3e7a1618	0.0.0.0:49153	svchost.exe
0x3e843b10	192.168.56.8:49177	svchost.exe
0x3e8f0620	0.0.0.0:49156	lsass.exe
0x3e9b3e50	0.0.0.0:49155	services.exe
0x3e1aa88	0.0.0.0:22	sshd.exe
0x3ee1bc18	0.0.0.0:22	sshd.exe
0x3fcaa008	192.168.56.8:49178	sshd.exe

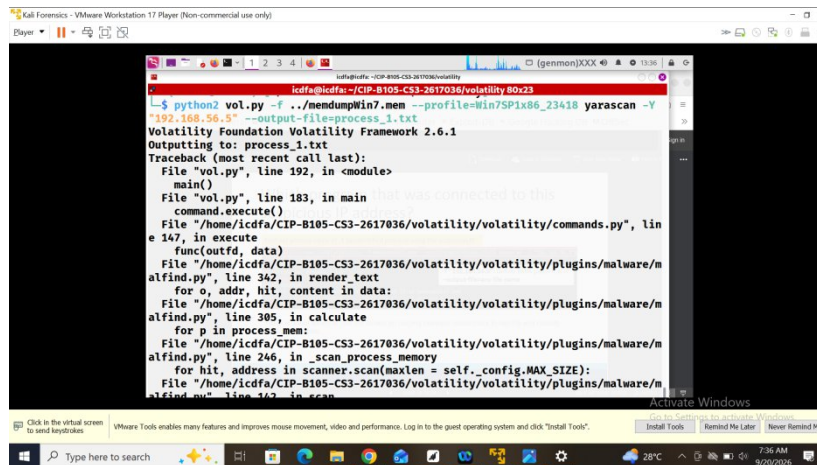
18. Program that was connected to the suspicious IP address



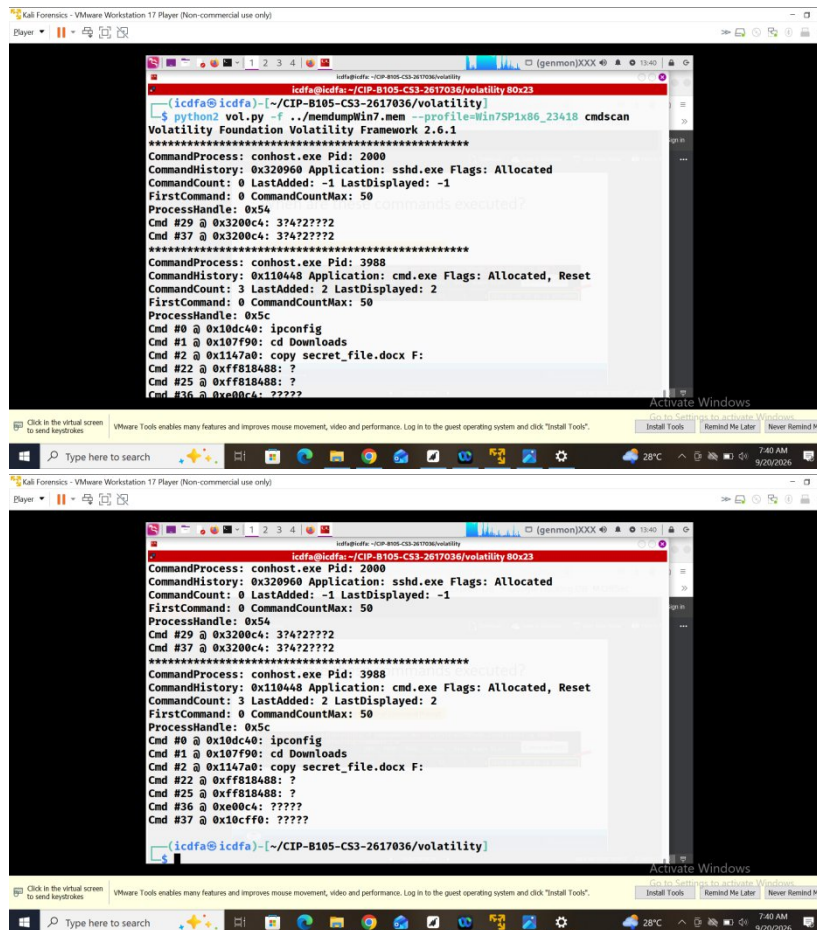
The screenshot shows the output of the command `python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86_23418 pslist -p 4`. It displays a list of processes with their PIDs, PPIDs, and other details.

PID	PPID	Process Name
0x8413a908	4	System
0x1061802	0	01-06 18:02:44 UTC+0000

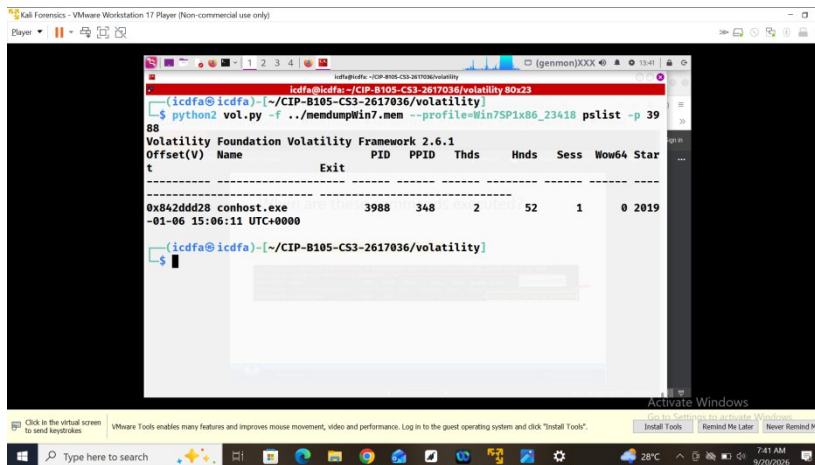
19. Finding process name using suspicious IP address



20. The suspect obviously used cmd to copy files



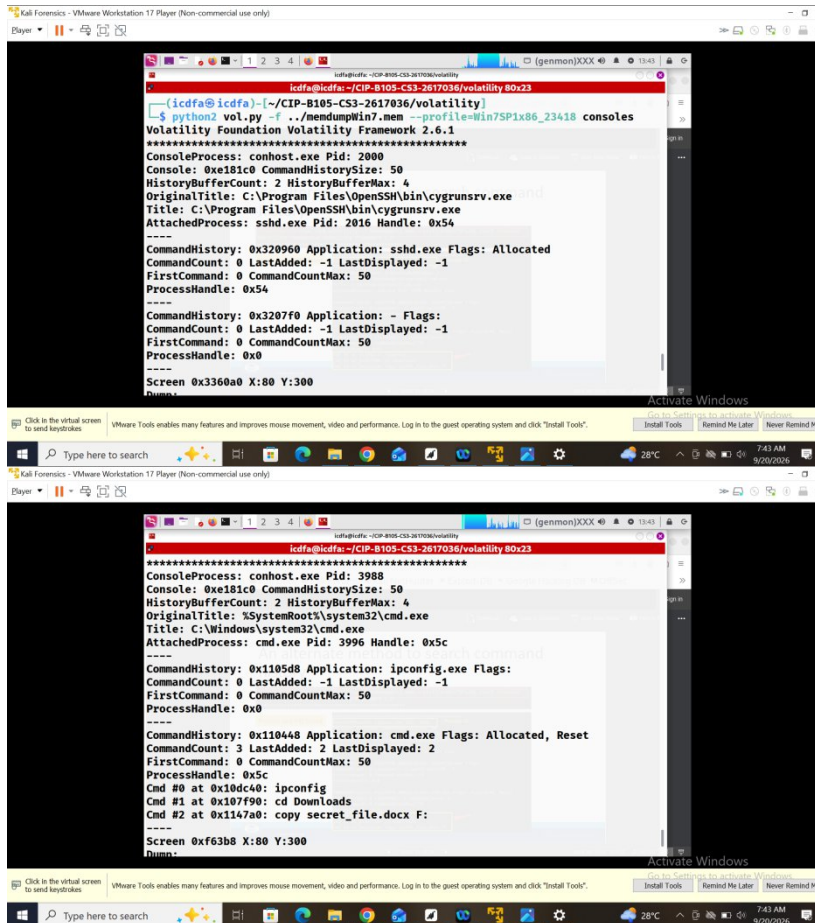
21. The timestamps the command were executed



```
icdfe@icdfe:~/CIP-B105-CS3-2617036/volatility 80x23
icdfe@icdfe:~/CIP-B105-CS3-2617036/volatility
$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86_23418 pslist -p 3988
Volatility Foundation Volatility Framework 2.6.1
Offset(V)  Name      Exit      PID  PPID  Thds  Hnds  Sess  Wow64  Start
-----
0x842ddd28  conhost.exe  3988  348  2     52    1    0  2019
-01-06 15:06:11 UTC+0000

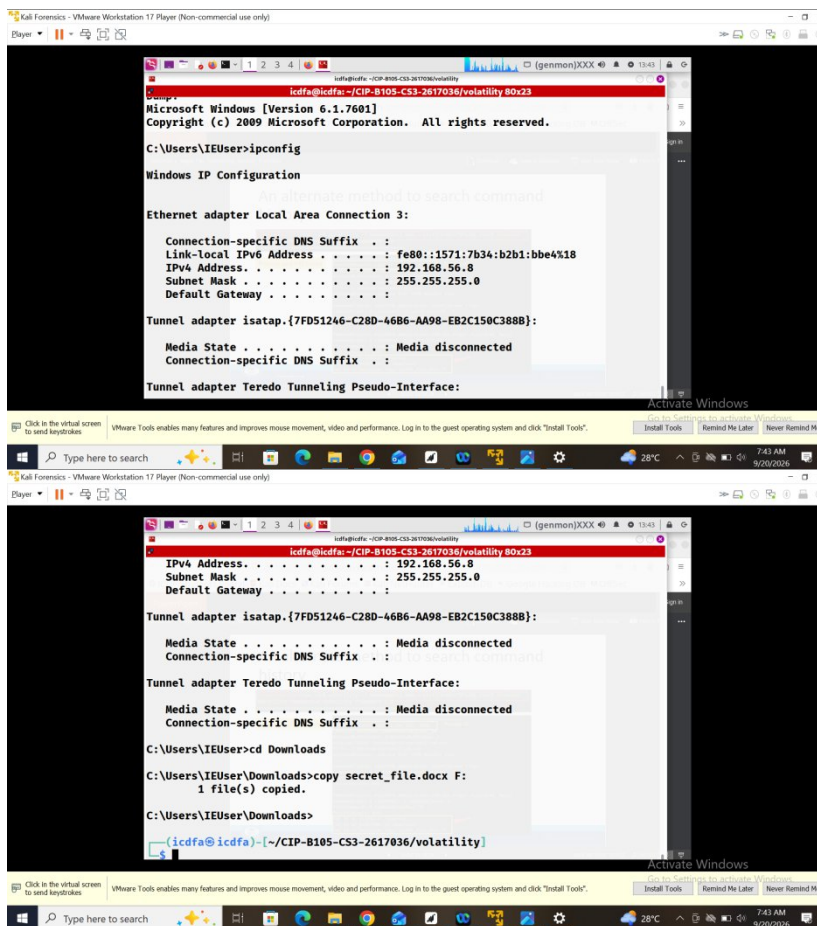
icdfe@icdfe:~/CIP-B105-CS3-2617036/volatility
$
```

22. Alternate method to search command history

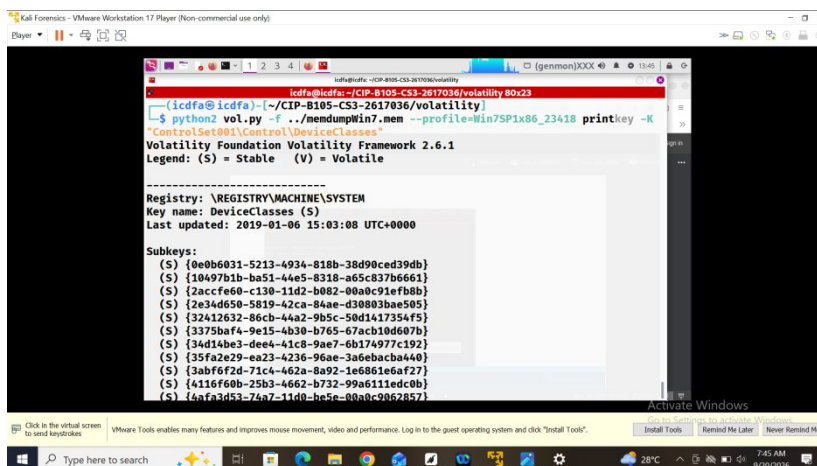


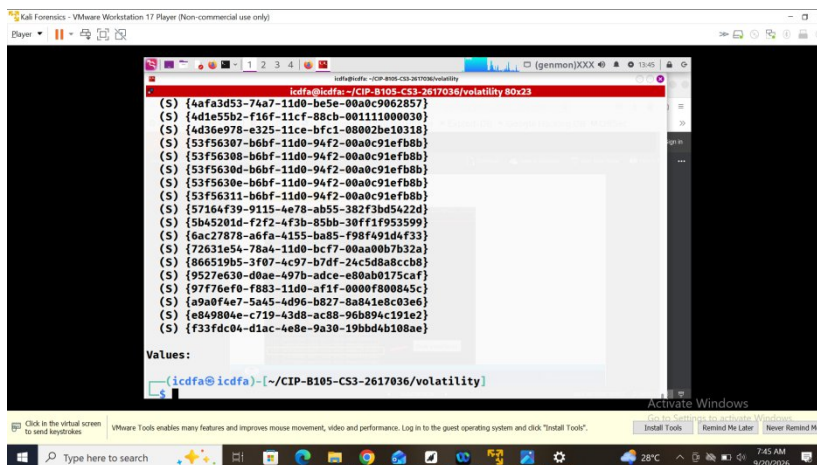
```
icdfe@icdfe:~/CIP-B105-CS3-2617036/volatility 80x23
icdfe@icdfe:~/CIP-B105-CS3-2617036/volatility
$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86_23418 consoles
Volatility Foundation Volatility Framework 2.6.1
*****
ConsoleProcess: conhost.exe Pid: 2000
Console: 0xe181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
Title: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
AttachedProcess: sshd.exe Pid: 2016 Handle: 0x54
-----
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
-----
CommandHistory: 0x3207f0 Application: - Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x0
-----
Screen 0x3360a0 X:80 Y:300
Dump:

icdfe@icdfe:~/CIP-B105-CS3-2617036/volatility 80x23
icdfe@icdfe:~/CIP-B105-CS3-2617036/volatility
$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86_23418 cmd
*****
ConsoleProcess: conhost.exe Pid: 3988
Console: 0xe181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: %SystemRoot%\system32\cmd.exe
Title: C:\Windows\system32\cmd.exe
AttachedProcess: cmd.exe Pid: 3996 Handle: 0x5c
-----
CommandHistory: 0x1105d8 Application: ipconfig.exe Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x0
-----
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 at 0x10dc40: ipconfig
Cmd #1 at 0x107f90: cd Downloads
Cmd #2 at 0x1147a0: copy secret_file.docx F:
-----
Screen 0xf63b8 X:80 Y:300
Dump:
```

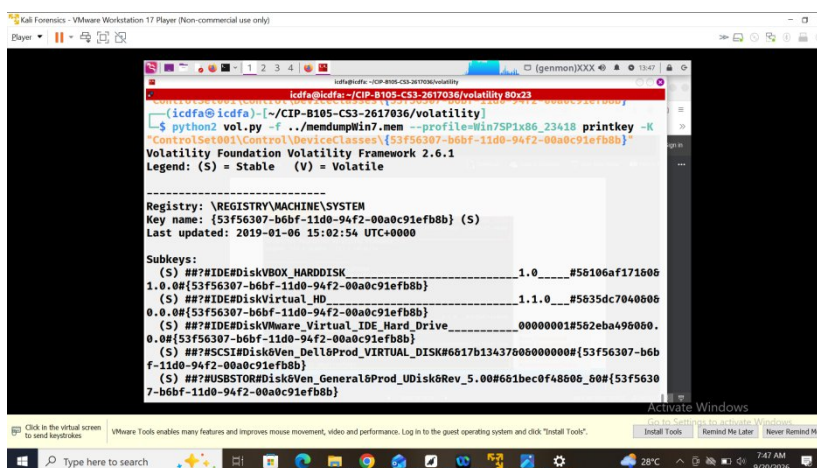


23. List of all device interfaces

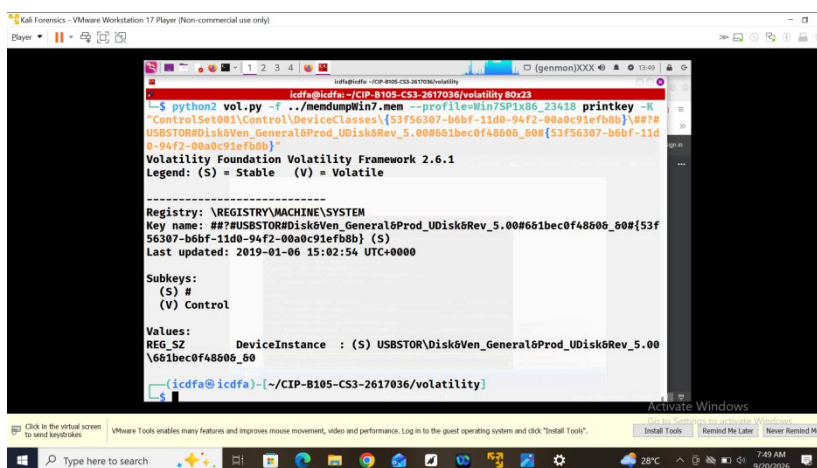




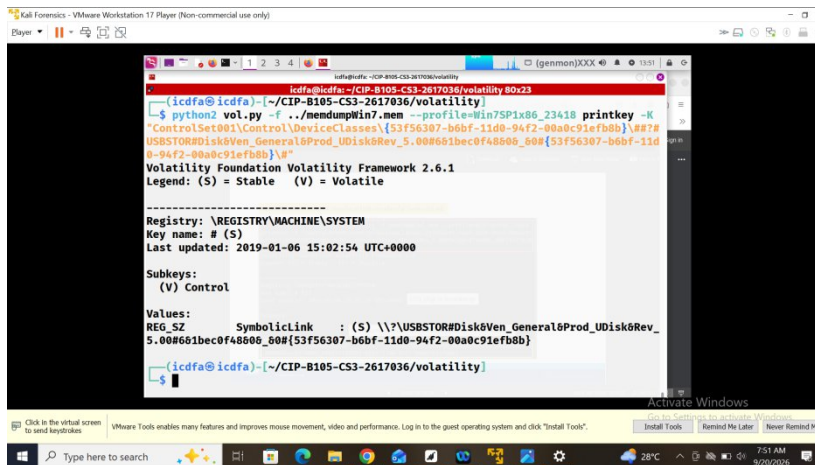
24. List of Disk Interfaces



25. Deviceclasses\Disk interface\USB interface



26. With symbolic link



```
icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility 80x23
icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility
$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K
"ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00a0c91fb8b}\#7#
USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#681bec0f4800_60#{53f56307-b6bf-11d0-94f2-00a0c91fb8b}"
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable (V) = Volatile

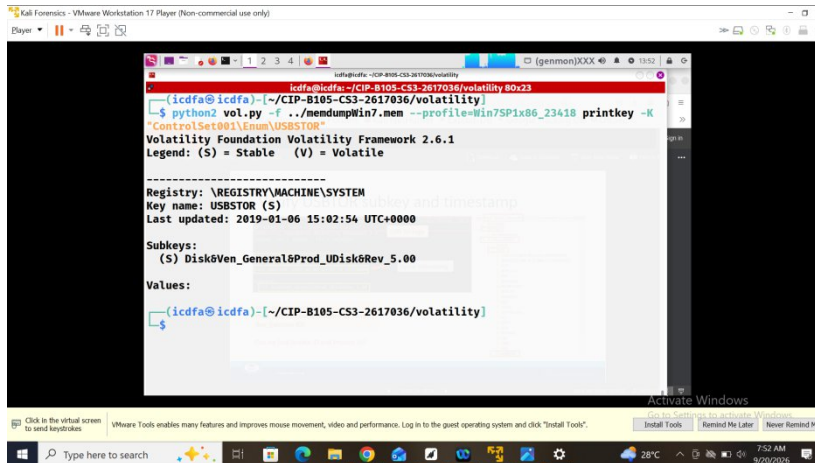
-----
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: # (S)
Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:
(V) Control

Values:
REG_SZ SymbolicLink : (S) \\?\USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_
5.00#681bec0f4800_60#{53f56307-b6bf-11d0-94f2-00a0c91fb8b}

icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility
$
```

27. Verification of USBTOR subkey and time stamp



```
icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility 80x23
icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility
$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K
"ControlSet001\Enum\USBSTOR"
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable (V) = Volatile

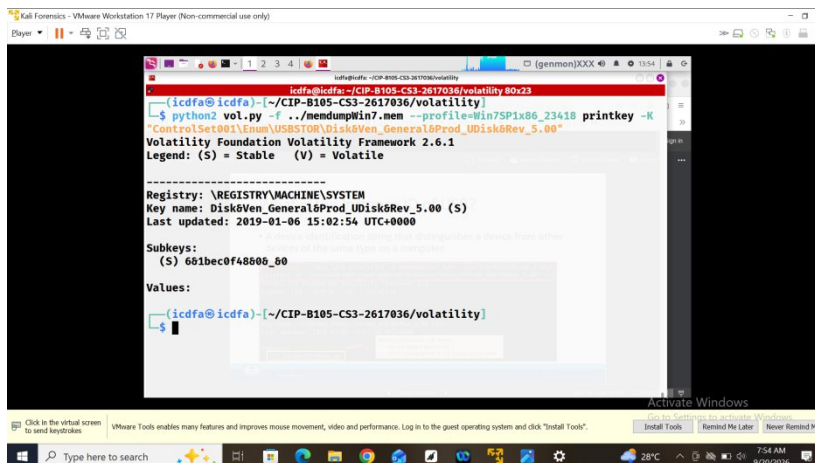
-----
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: USBSTOR (S)
Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:
(S) Disk&Ven_General&Prod_UDisk&Rev_5.00

Values:

icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility
$
```

28. The instance ID of the USB



```
icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility 80x23
icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility
$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K
"ControlSet001\Enum\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00"
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable (V) = Volatile

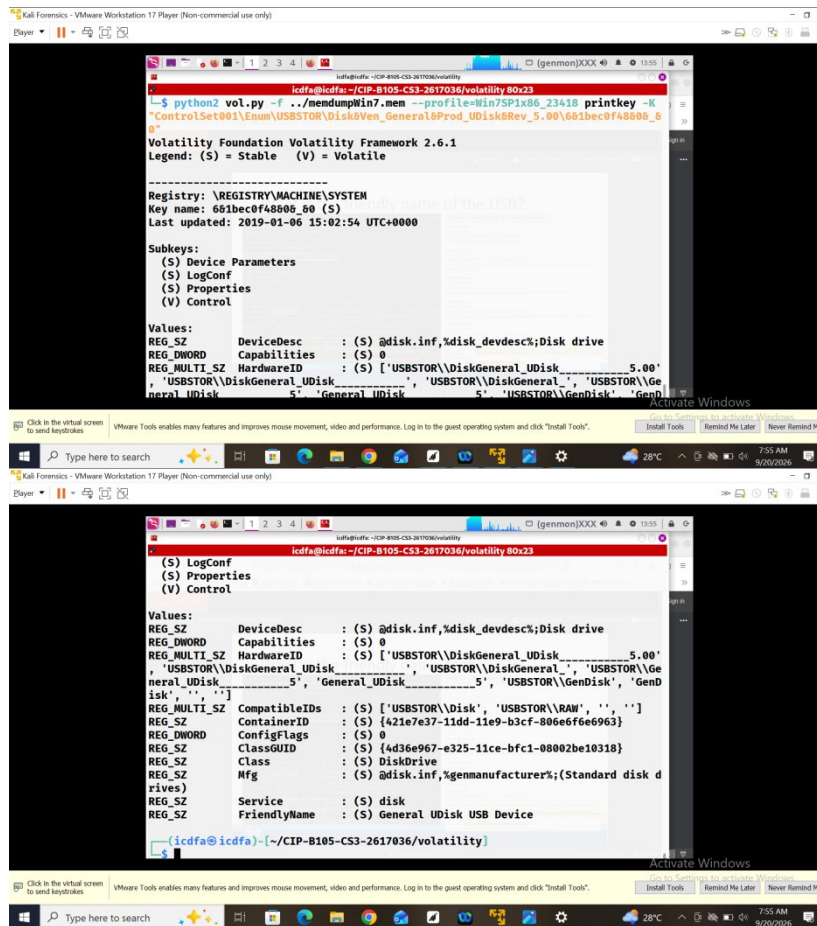
-----
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: Disk&Ven_General&Prod_UDisk&Rev_5.00 (S)
Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:
(S) 681bec0f4800_60

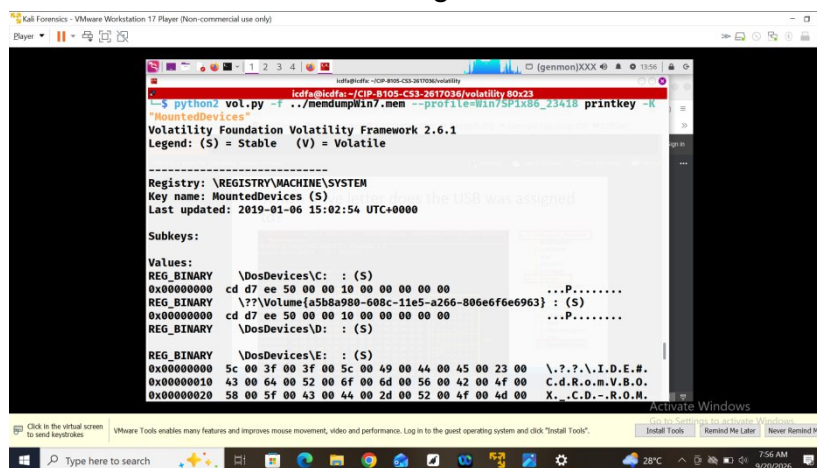
Values:

icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility
$
```

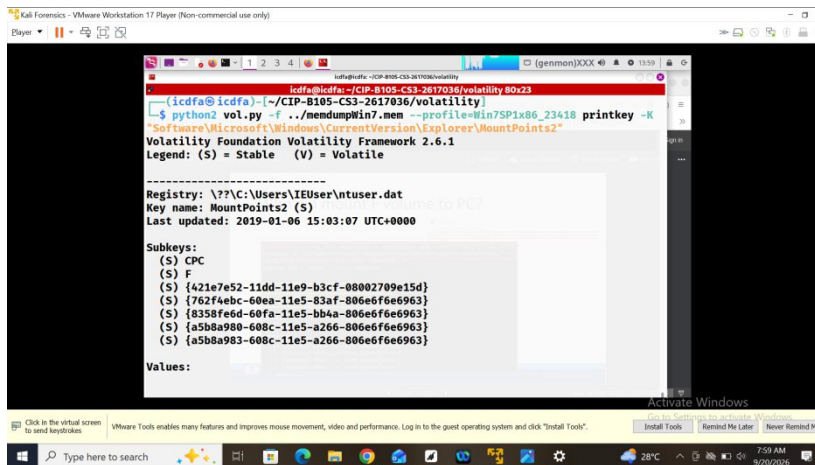
29. The friendly name of the USB



30. The drive letter the USB was assigned to



31. The mount F volume to PC



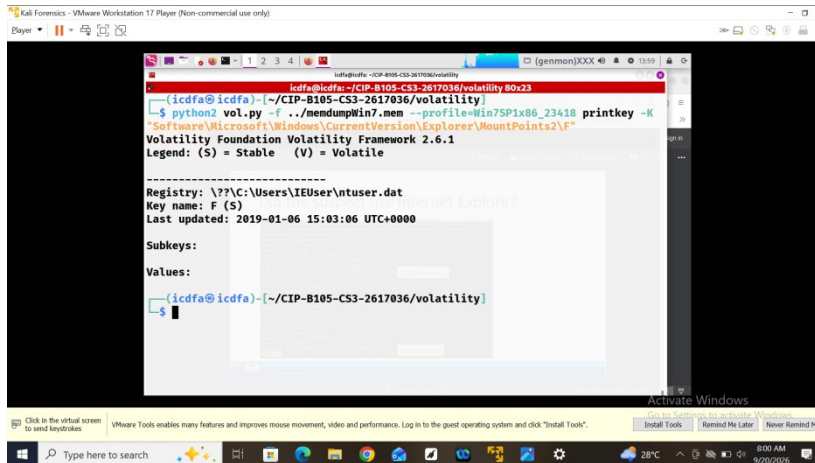
```
icdfe@icdfe:~/CIP-B105-CS3-2617036/volatility 80x23
icdfe@icdfe:~/CIP-B105-CS3-2617036/volatility
$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K
"Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2"
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable (V) = Volatile

-----
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: MountPoints2 (S)
Last updated: 2019-01-06 15:03:07 UTC+0000

Subkeys:
(S) CPC
(S) F
(S) {421e7e52-11dd-11e9-b3cf-08002709e15d}
(S) {762f4ebc-60ea-11e5-83af-806e6f6e6963}
(S) {8358fe6d-60fa-11e5-bb4a-806e6f6e6963}
(S) {a5b8a980-608c-11e5-a266-806e6f6e6963}
(S) {a5b8a983-608c-11e5-a266-806e6f6e6963}

Values:
```

32. When the USB was last attached to PC

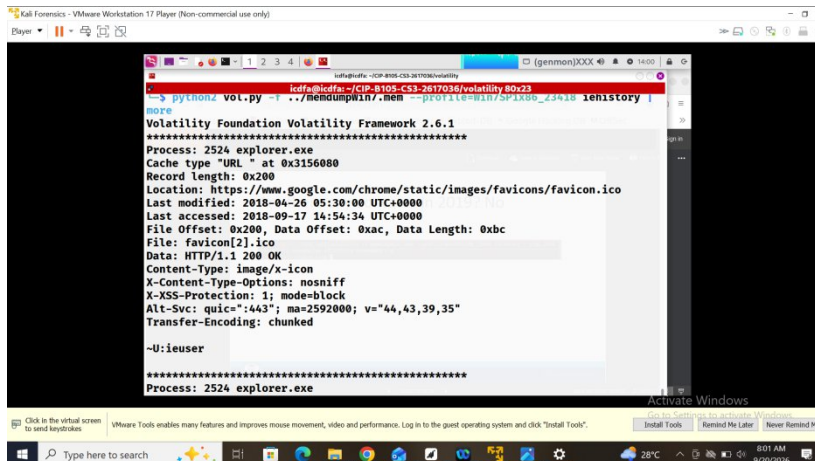


```
icdfe@icdfe:~/CIP-B105-CS3-2617036/volatility 80x23
icdfe@icdfe:~/CIP-B105-CS3-2617036/volatility
$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K
"Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2\F"
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable (V) = Volatile

-----
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: F (S)
Last updated: 2019-01-06 15:03:06 UTC+0000

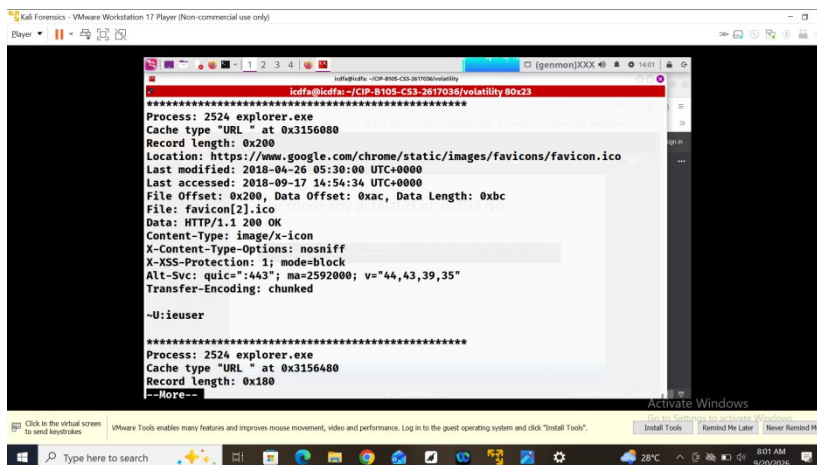
Subkeys:
Values:
icdfe@icdfe:~/CIP-B105-CS3-2617036/volatility
$
```

33. The suspect then used internet Explorer

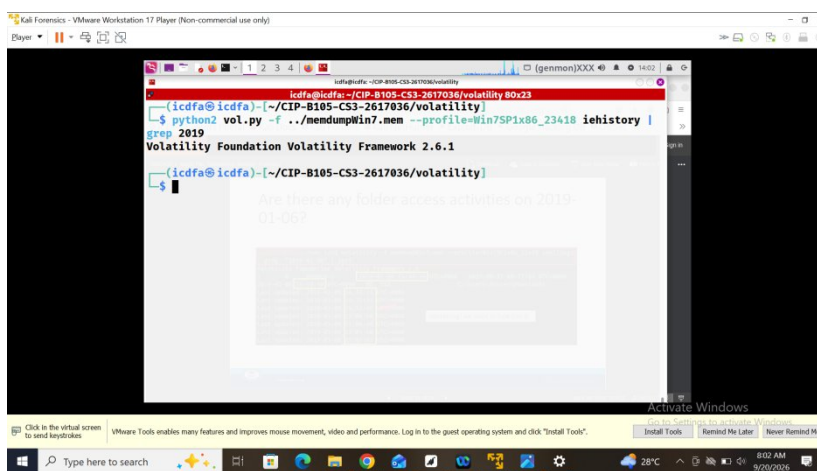


```
icdfe@icdfe:~/CIP-B105-CS3-2617036/volatility 80x23
icdfe@icdfe:~/CIP-B105-CS3-2617036/volatility
$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86_23418 iehistory |
more
Volatility Foundation Volatility Framework 2.6.1
*****
Process: 2524 explorer.exe
Cache type "URL " at 0x3156080
Record length: 0x200
Location: https://www.google.com/chrome/static/images/favicons/favicon.ico
Last modified: 2018-04-26 05:30:00 UTC+0000
Last accessed: 2018-09-17 14:54:34 UTC+0000
File Offset: 0x200, Data Offset: 0xac, Data Length: 0xbc
File: favicon[2].ico
Data: HTTP/1.1 200 OK
Content-Type: image/x-icon
X-Content-Type-Options: nosniff
X-XSS-Protection: 1; mode=block
Alt-Svc: quic=":443"; ma=2592000; v="44,43,39,35"
Transfer-Encoding: chunked

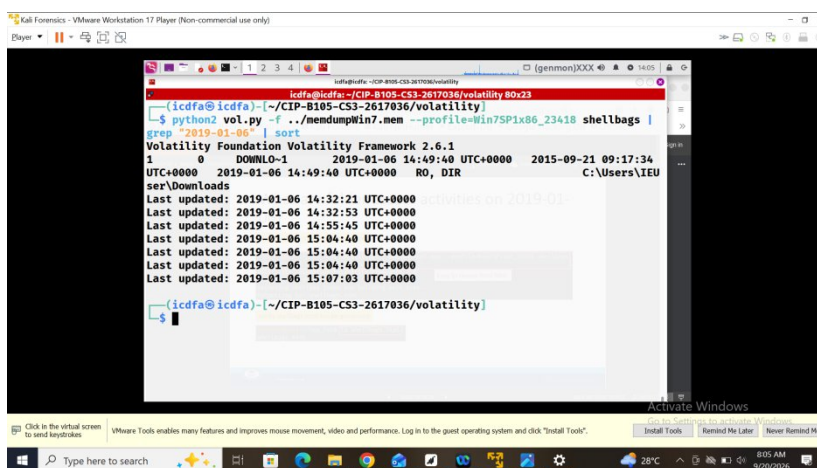
~U:ieuser
*****
Process: 2524 explorer.exe
```



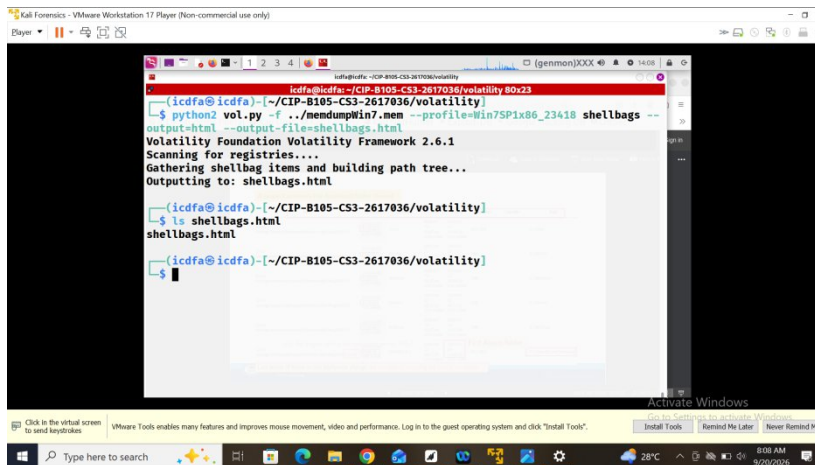
34. There were no activates in 2019



35. Folder access activities on 2019-01-06



36. Printing shellbags to html extension and verifying it

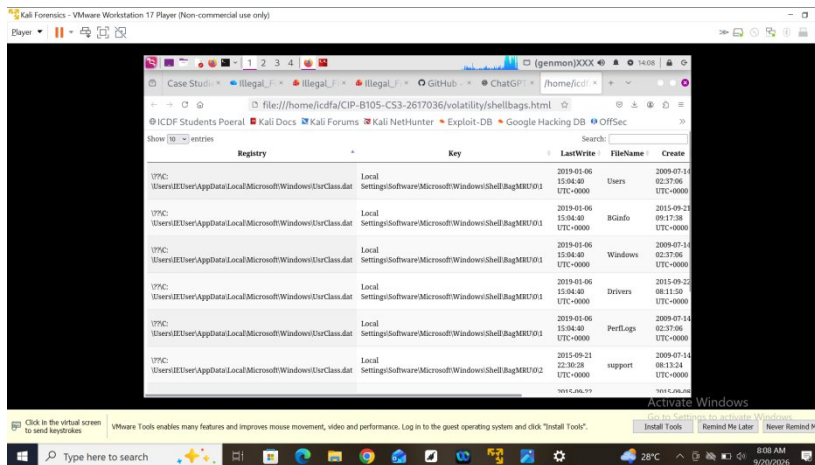


```
icdffa@icdffa:~/CIP-B105-CS3-2617036/volatility 80x23
icdffa@icdffa:~/CIP-B105-CS3-2617036/volatility
$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86_23418 shellbags --output=html --output-file=shellbags.html
Volatility Foundation Volatility Framework 2.6.1
Scanning for registries....
Gathering shellbag items and building path tree...
Outputting to: shellbags.html

icdffa@icdffa:~/CIP-B105-CS3-2617036/volatility
$ ls shellbags.html
shellbags.html

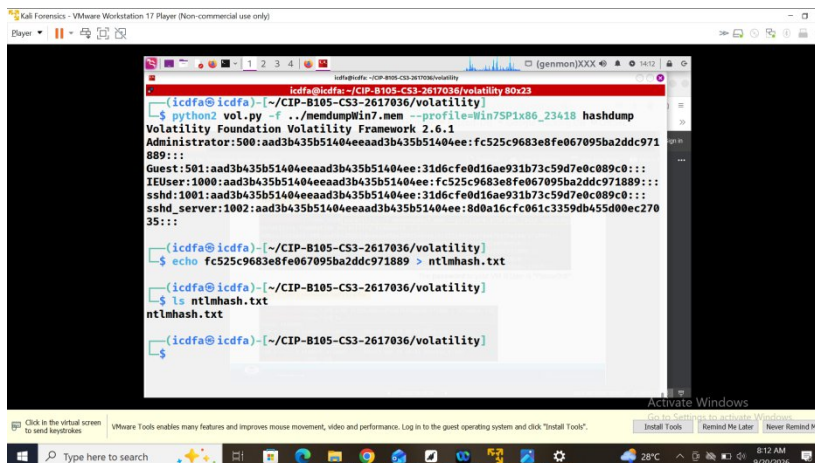
icdffa@icdffa:~/CIP-B105-CS3-2617036/volatility
$
```

37. The evidence



Registry	Key	LastWrite	FileName	Create
U79C: (Users)IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU01	2019-01-06 15:04:40 UTC-0000	Users	2009-07-10 02:37:06 UTC-0000
U79C: (Users)IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU01	2019-01-06 15:04:40 UTC-0000	BGInfo	2015-09-25 09:17:38 UTC-0000
U79C: (Users)IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU01	2019-01-06 15:04:40 UTC-0000	Windows	2009-07-10 02:37:06 UTC-0000
U79C: (Users)IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU01	2019-01-06 15:04:40 UTC-0000	Drivers	2015-09-25 08:11:59 UTC-0000
U79C: (Users)IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU01	2019-01-06 15:04:40 UTC-0000	PerfLogs	2009-07-10 02:37:06 UTC-0000
U79C: (Users)IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU02	2015-09-21 22:30:28 UTC-0000	support	2009-07-14 08:13:24 UTC-0000

38. Dump NTLM hash



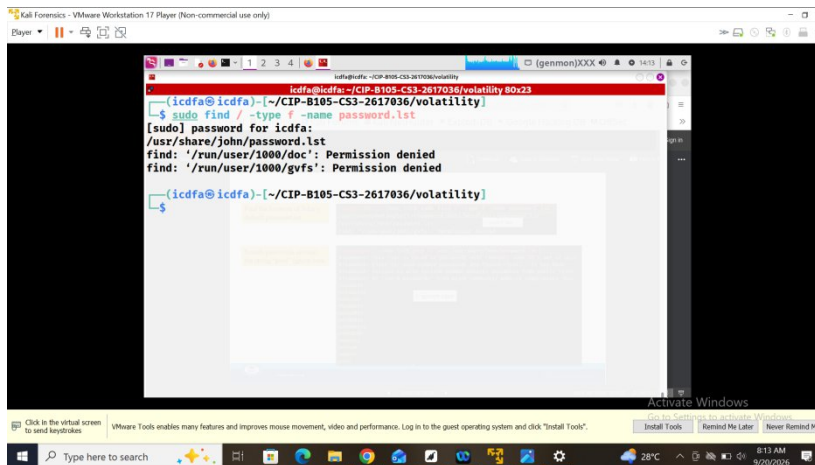
```
icdffa@icdffa:~/CIP-B105-CS3-2617036/volatility 80x23
icdffa@icdffa:~/CIP-B105-CS3-2617036/volatility
$ python2 vol.py -f ../memdumpWin7.mem --profile=Win7SP1x86_23418 hashdump
Volatility Foundation Volatility Framework 2.6.1
Administrator:500:aad3b435b51404eeaad3b435b51404ee:fc525c9683e8fe067095ba2ddc97f1889:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
IEUser:1000:aad3b435b51404eeaad3b435b51404ee:fc525c9683e8fe067095ba2ddc97f1889:::
sshd:1001:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
sshd_server:1002:aad3b435b51404eeaad3b435b51404ee:8d0a16cfc061c359db455d00ec27035:::

icdffa@icdffa:~/CIP-B105-CS3-2617036/volatility
$ echo fc525c9683e8fe067095ba2ddc97f1889 > ntlmhash.txt

icdffa@icdffa:~/CIP-B105-CS3-2617036/volatility
$ ls ntlmhash.txt
ntlmhash.txt

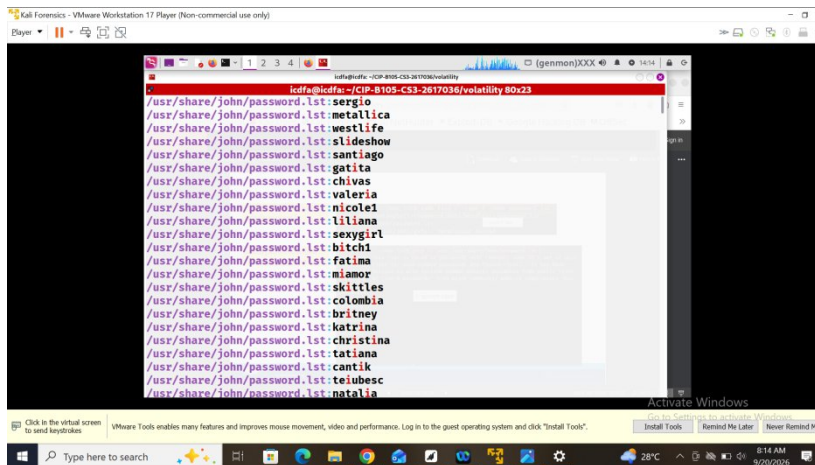
icdffa@icdffa:~/CIP-B105-CS3-2617036/volatility
$
```

39. The location of john's default password.lst



```
icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility 80x23
[icdfa@icdfa]~/CIP-B105-CS3-2617036/volatility
$ sudo find / -type f -name password.lst
[sudo] password for icdfa:
/usr/share/john/password.lst
find: '/run/user/1000/doc': Permission denied
find: '/run/user/1000/gvfs': Permission denied
[icdfa@icdfa]~/CIP-B105-CS3-2617036/volatility
$
```

40. Searching password that contains the string “pass” and ignoring the case



```
icdfa@icdfa:~/CIP-B105-CS3-2617036/volatility 80x23
/usr/share/john/password.lst:sergio
/usr/share/john/password.lst:metallica
/usr/share/john/password.lst:westlife
/usr/share/john/password.lst:slideshow
/usr/share/john/password.lst:santiago
/usr/share/john/password.lst:gatita
/usr/share/john/password.lst:chivas
/usr/share/john/password.lst:valeria
/usr/share/john/password.lst:nicole1
/usr/share/john/password.lst:liliana
/usr/share/john/password.lst:sexygirl
/usr/share/john/password.lst:bitch1
/usr/share/john/password.lst:fatima
/usr/share/john/password.lst:miamor
/usr/share/john/password.lst:skittles
/usr/share/john/password.lst:colombia
/usr/share/john/password.lst:britney
/usr/share/john/password.lst:katrina
/usr/share/john/password.lst:christina
/usr/share/john/password.lst:tatiana
/usr/share/john/password.lst:cantik
/usr/share/john/password.lst:teibesc
/usr/share/john/password.lst:netalin
```

Conclusion

The lab successfully demonstrated how Volatility 2 can be used to correlate memory, Registry, browser, command-history, Shellbag, and USB artefacts. The combined evidence established a consistent sequence of downloading a file and transferring it to removable storage, with the relevant user, device, drive letter, and timestamps corroborated across multiple artefacts.